

MONEY.exe
NOT FOUND

SMART INDIA HACKATHON 2026



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HACKATHON
2026

TITLE PAGE

- Problem Statement ID – **SIH26172**
- Problem Statement Title- **Low Latency and Efficient Voice Activator for Edge Devices**
- Theme- **Smart Automation**
- PS Category- **Hardware**
- Team ID- **177073**
- Team Name- **MONEY.exe NOT FOUND**



TORS

The Egde That Listens.





CHALLENGES

- Cloud-centric voice adds latency
- High bandwidth and cost
- Privacy and data exposure

CURRENT LIMITATIONS

- Continuous streaming wastes power
- Network dependency adds delay
- MCU constraints limit on-device models

EXISTING OPTIONS

- Cloud-first voice assistants
- Proprietary wake-word SDKs
- Generic pre-trained keywords

OUR APPROACH

ESP32-S3 + INMP441

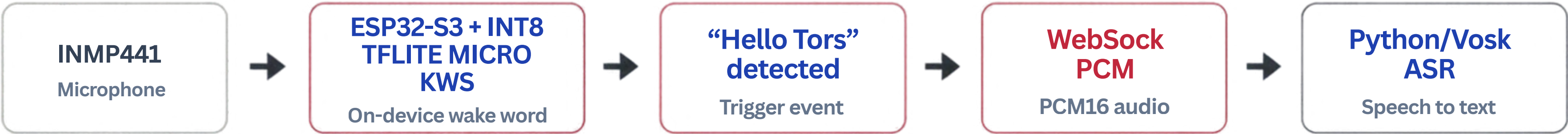
Custom “Hello Tors” KWS
INT8 TFLite Micro + mfCC

~800 ms pre-trigger buffer

Trigger-gated PCM16
WebSocket to ASR

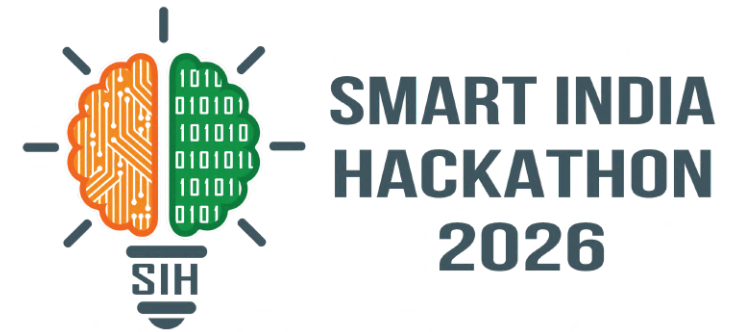
CHALLENGE CONSTRAINTS	<256 KB RAM	<10%IDLE CPU	ACCURACY	LATENCY
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Wake-word: “Hello Tors.”

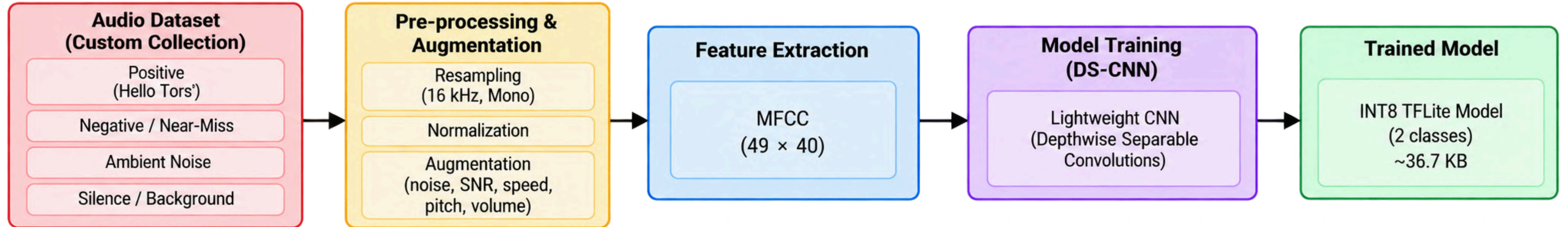


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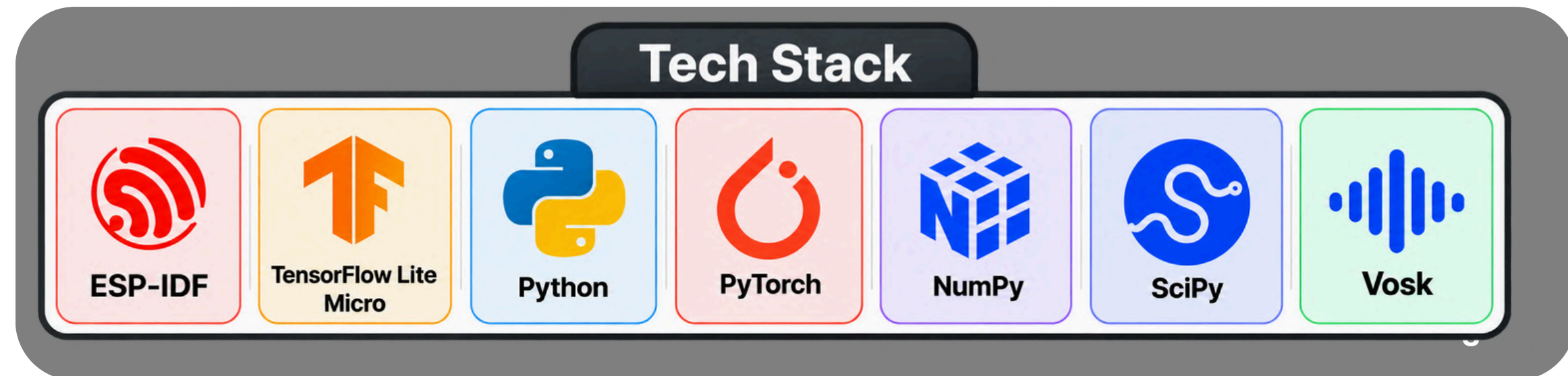
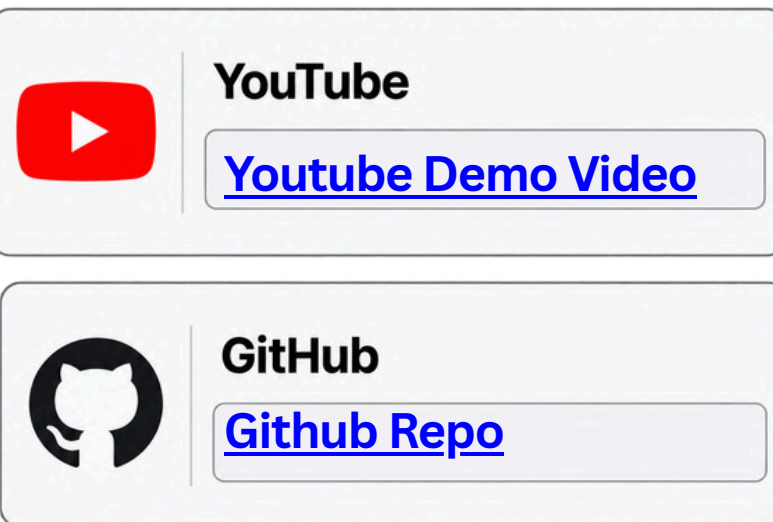
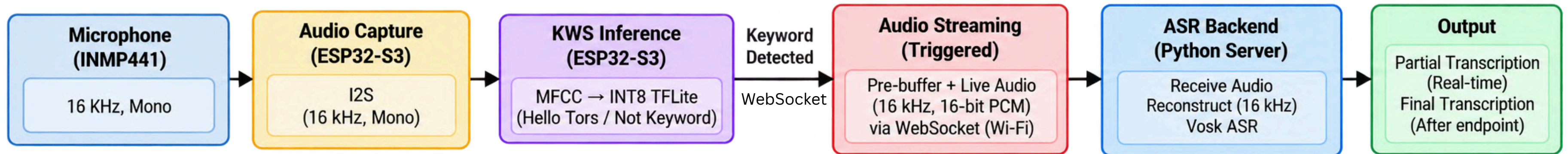
TECHNICAL APPROACH



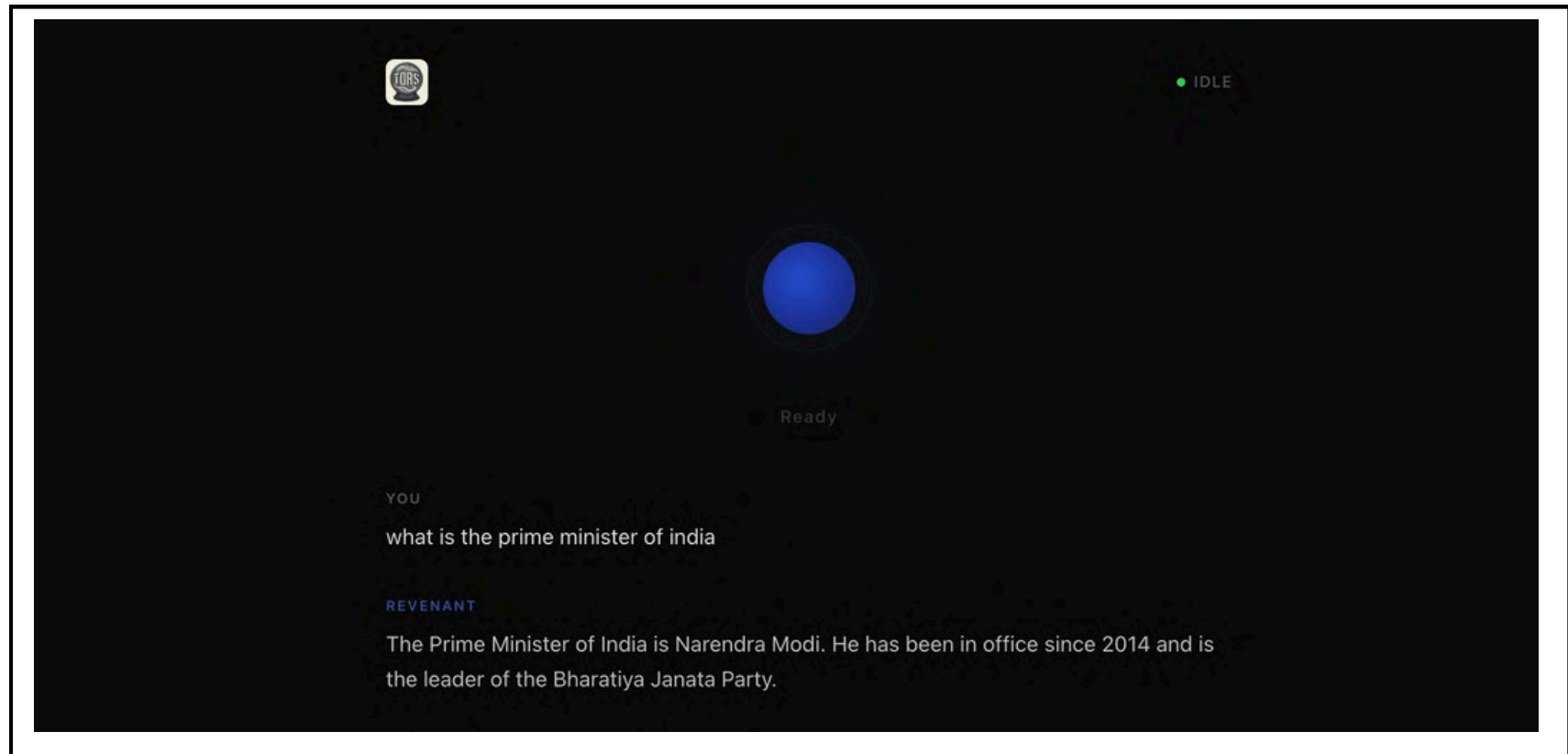
1. Model Training (Offline)



2. Real-Time Inference and Audio Streaming (Deployment)

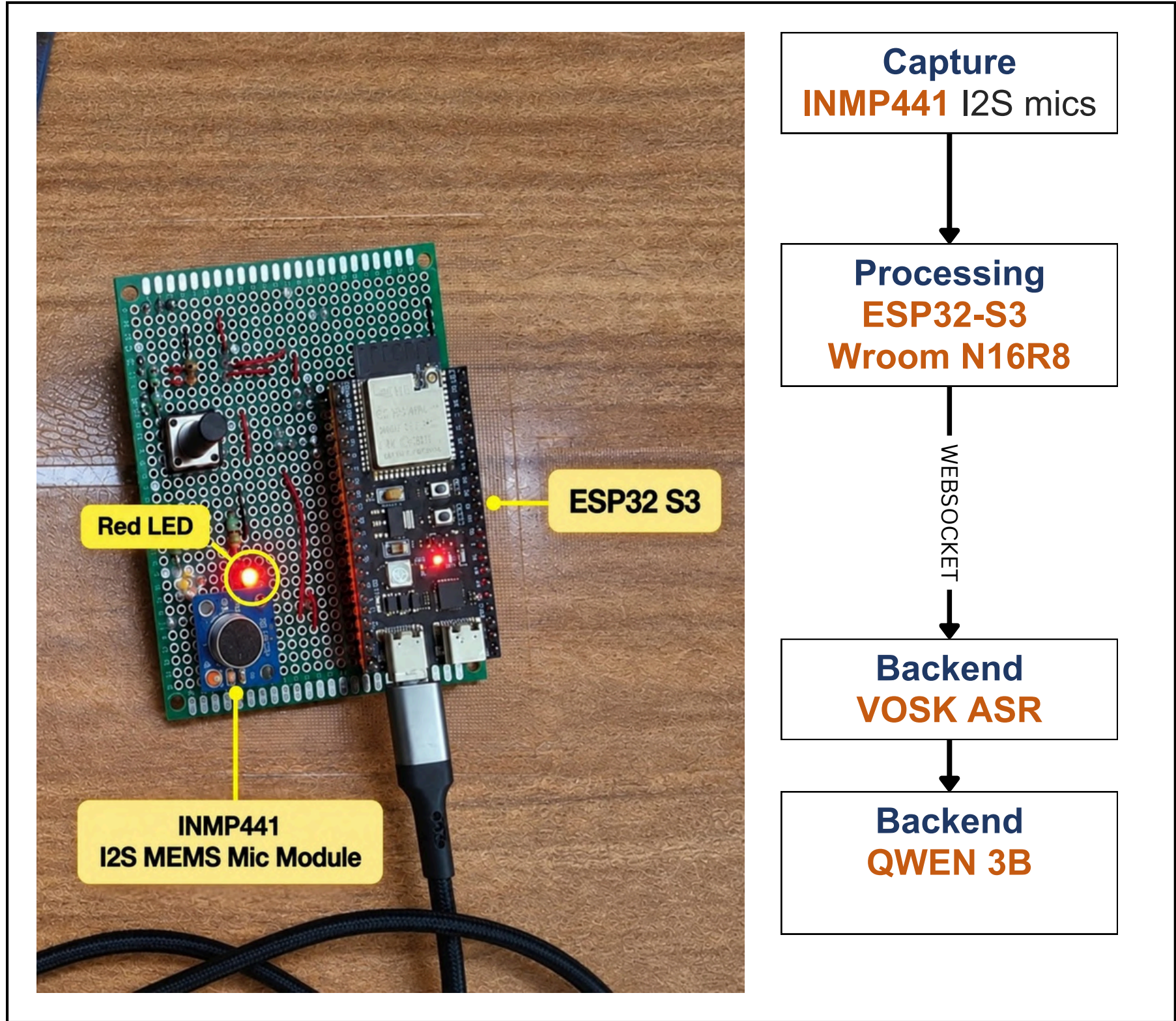


DASHBOARD



Bill of Materials (BOM)

S. No.	Component	Quantity	Unit Price (INR)	Total Price (INR)
1	ESP32-S3 WROOM-1 N16R8	1	720	720
2	INMP441 Microphone	1	349	349
3	Wiring and Misc	1	45	45
Total (INR)				1,114



Hardware Prototype

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FEASIBILITY AND VIABILITY

Capability / Requirement	 Cloud Voice Assistants e.g. Alexa, Google Assistant	 Proprietary Wake-Word SDKs e.g. Picovoice, Sensory	 TORS (Our Solution) ESP32-S3 + Custom KWS
Local keyword detection	✗	✓	✓
Custom keyword support	△	✓	✓
Open-source stack	✗	✗	✓
Runs on low-power MCU	△	✓	✓
Continuous cloud audio required	✓	✓	✗
Trigger-gated audio streaming	✗	△	✓
Customizable KWS model	△	△	✓
Edge + server architecture	✓	✓	✓
Designed for < 256 KB RAM constraint	✗	△	✓ (Target specification)
No commercial SDK dependency	✗	✗	✓

- ✓

= Supported
- ✗

= Not supported
- △

= Partially supported / Depends



Feasible
Built using low-cost, widely available hardware and open-source tools



Viable
Meets the key requirements of low-resource, always-on operation with customizable keywords

Latency Breakdown

Latency Metric

Approx. Value


Model inference

121 ms


Keyword detection

0.3–0.7 s


Wi-Fi delivery

5–30 ms


Wake-to-server

0.3–0.75 s

 KWS subsystem RAM, estimated	127–137 KB	✓ Below 256 KB when considered by itself; not independently measured
 Full firmware internal RAM, approximate	350 KB 160 KB static data/BSS + 190 KB allocated heap	 After optimization: < 10% average CPU utilization across both cores during continuous listening

KEY NUMBERS

~2.7 GB/DAY AUDIO KEPT ON-DEVICE

0 B UPLOADED WHILE IDLE

36.7 KB MODEL

~2.4% PACKET OVERHEAD

| BENEFITS — FOR USERS & OPERATORS

PRIVACY

Privacy by design

Audio stays on the ESP32 until “Hello Tors”; zero ambient audio is uploaded.

COST

Lower cloud & bandwidth bills

Avoids ~2.7 GB/day of raw audio per device; no work on silence.

SPEED

Faster response

Warm WebSocket + 800 ms pre-buffer: no reconnect delay, no clipped words.

HARDWARE

Runs on low-cost MCUs

36.7 KB INT8 model on ESP32-S3 — no GPU, no Linux board, battery-friendly.

RELIABILITY

Fewer false wake-ups

2-window trigger rule + 1 s refractory; wake-up works even on weak networks.

| IMPACT — SOCIETY & NATION

1

HANDS-FREE OPS

Voice control for labs, control rooms and field sites where hands are busy.

2

DIGITAL INCLUSION

Local wake-up works on patchy networks: rural homes, farms, disaster zones.

3

ATMANIRBHAR & OPEN

Open-source stack, no licensed SDKs; keywords in Indian languages.

4

GREEN AT SCALE

Idle devices use near-zero data, compute and energy — scales to millions.

RECENT RESEARCH

2026

Optimizing Keyword Spotting Classifier Based on Tiny Machine Learning for Low-Power Embedded Devices

2025 MIPRO 48th ICT and Electronics Convention · IEEE
ieeexplore.ieee.org/document/11131906

2025

A KWS System for Edge-Computing Applications with Analog-Based Feature Extraction and Learned Step Size Quantized Classifier

Sensors, Vol. 25, No. 8, 2550 · 2025
mdpi.com/1424-8220/25/8/2550

2024

Trade-Offs in Neural Network Compression: Quantized and Binary Models for Keyword Spotting

2024 IEEE International Conference on Electronics, Circuits and Systems (ICECS) · 2024
ieeexplore.ieee.org/document/10849192

2024

On-Device Domain Learning for Keyword Spotting on Low-Power Extreme Edge Embedded Systems

C. Cioflan et al. · AICAS 2024
ieeexplore.ieee.org/document/10595987

2024

Exploring TinyML Frameworks for Small-Footprint Keyword Spotting: A Concise Overview

S. Garai and S. Samui · SPCOM 2024
ieeexplore.ieee.org/document/10631631



DATASET & IMPLEMENTATION BASIS



CUSTOM DATASET

Team-recorded real-world audio

Positive · Negative/Near-miss ·
Ambient noise · Silence/background



TRAINING DATA POLICY

No open-source audio used for training



DEPLOYMENT BASIS

ESP32-S3 · INMP441 · 16 KHz mono
49 × 40 MFCC · INT8 TFLite Micro · 2-class KWS